



مدارس حديد عز
للتكنولوجيا التطبيقية
ezzsteel applied technology schools

Subtopic 3.1 — The EU Workplace Safety Framework

CE Marking & Key EU Product Directives

Eng. /Nada Ahmed Ali Attia

Date:- 7 / 10 / 2026

Academic Year: 2026/2027
Semester : One

What is the first thing you see in the photo?



Guide them throw
identification plate
And C E

THE EU WORKPLACE SAFETY FRAMEWORK

European safety legislation helps protect both

1- workers

2- users of products and equipment.

It includes requirements for safe working conditions as well as requirements that apply to products placed on the market

Let's compare between them



THE EU WORKPLACE SAFETY FRAMEWORK

comparison

spect	Workplace-Safety Directives	Product-Safety Directives
Focus	Workers and working conditions	Products placed on the market
Main Target	Employers and workplace operators	Manufacturers and economic operators
Requirements	Risk identification, assessment, & control	Design, EHSR compliance, & conformity
Core Actions	Prevention, protection, info, & training	Documentation, assessment, & CE marking

Crucial Rule: *CE marking on a machine does not eliminate the employer's duty to provide training, procedures, and local safety controls.*



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Subtopic 3.2 – FROM EU PRODUCT DIRECTIVES TO SAFE PRODUCT DESIGN

CE Marking & Key EU Product Directives

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Product Directives & Harmonised Standards

An important approach to European product regulation is based on two connected parts:

Product Directives

Set mandatory Essential Health and Safety Requirements (EHSRs) — defining *WHAT* safety level must be achieved.

Harmonised Technical Standards

Provide technical specifications and risk assessment engineering methods — defining *HOW* requirements can be met.

ESSENTIAL HEALTH AND SAFETY REQUIREMENTS

Definition - EHSRs

Essential Health and Safety Requirements (EHSRs) are the minimum health and safety outcomes that a product covered by the relevant directive must satisfy

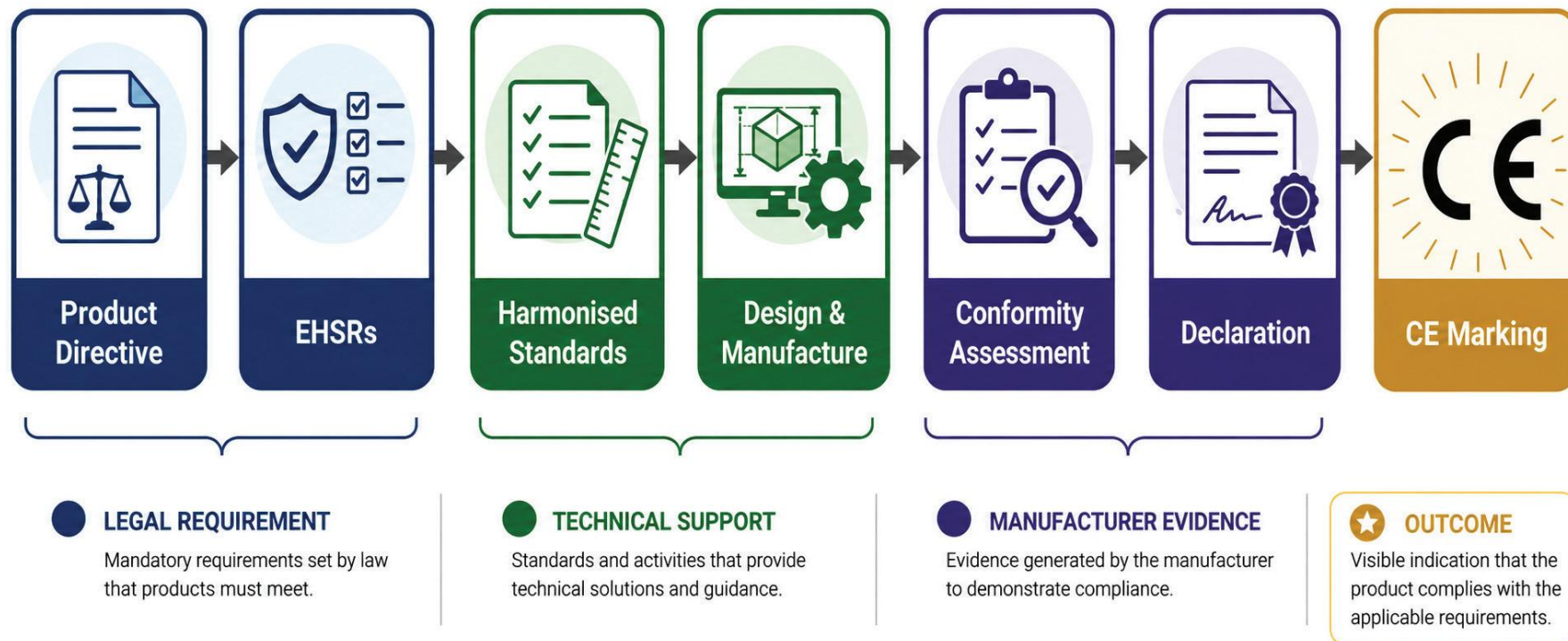
The 7-Step Conformity Assessment Pathway

From EU Directives to Safe Product Design

The 7 Sequential Steps:



EUROPEAN DIRECTIVES AND TECHNICAL SAFETY STANDARDS



Homework & Class Exercises

Part A: Multiple Choice Questions

What is the primary purpose of Council Directive 89/391/EEC (Framework Directive)?

- A) To regulate the pricing of industrial goods in Europe
- B) To establish a general framework for protecting workers' health and safety at work
- C) To eliminate the need for CE marking on industrial machines

Who holds primary responsibility under EU Product-Safety Directives?

- A) The Supervisor on the factory floor
- B) The Occupational Health Doctor
- C) The Manufacturer and economic operators placing the product on the market

What is the relationship between a Product Directive and a Harmonised Technical Standard (e.g., EN ISO 12100)?

- A) Directives specify *what* safety outcomes are required; standards provide technical methods on *how* to achieve them.
- B) Directives apply to workers; standards apply only to employers.
- C) Standards replace the legal requirement of the directive.

Does buying a CE-marked machine mean no further safety measures are needed in the workplace?

- A) Yes, CE marking guarantees that no training or local procedures are required.
- B) No, workplace safety controls, operator training, and safe work procedures are still required.
- C) Yes, because CE marking legally transfers all employer liability to the manufacturer.



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Subtopic 3.3 — European Directives and Technical Safety Standards

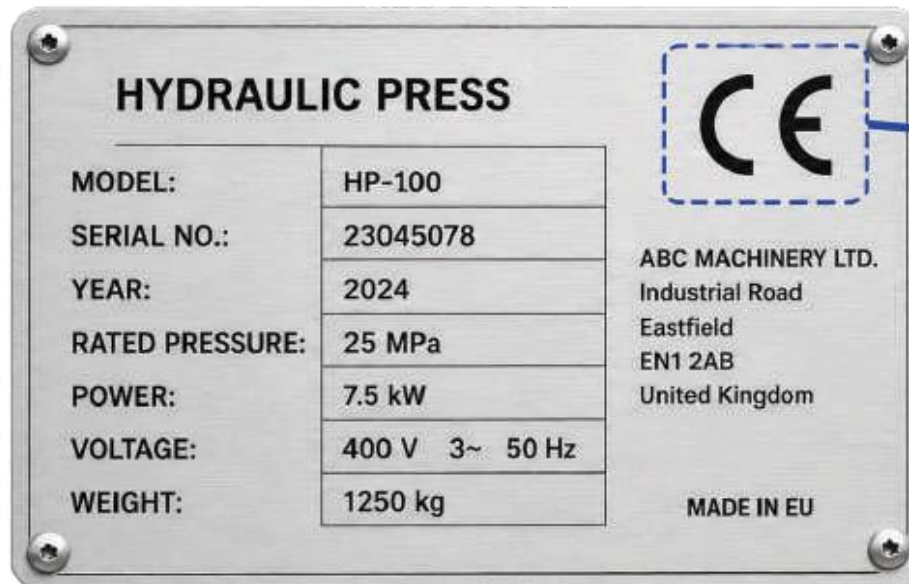
CE Marking & Key EU Product Directives

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What is this ?



Conformity Mark

What Does CE Marking Mean ?

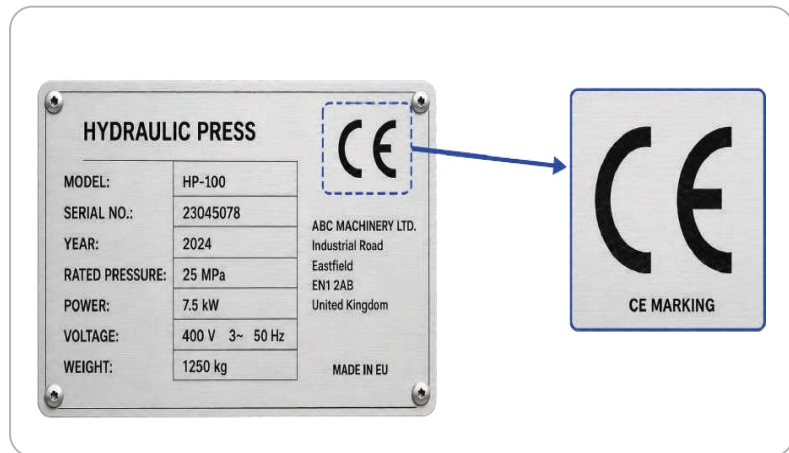
You will see the CE marking on many types of industrial equipment



Definition - CE MARKING

The CE marking is placed on a product by the manufacturer to declare that the applicable EU requirements have been addressed and that the required conformity-assessment procedures have been completed.

- ✓ **Not a quality mark**
CE is a conformity marking — not a general quality mark.
- ✓ **A defined form**
Follows a specified geometric form, with a minimum height of 5 mm.
- ✓ **Identifies the manufacturer**
Associated with the manufacturer or authorised representative.



A CE marking on an equipment nameplate

What CE Marking Does Not Tell You :-

Seeing a CE marking does not remove your responsibility to work safely



A marking is one part of the safety information available to you.

Before using industrial equipment , you may still need to consider:

- ✓ Whether you are looking at the correct equipment
- ✓ Whether the equipment is in suitable condition
- ✓ Whether guards or protective devices are present
- ✓ Whether you have the correct instructions
- ✓ Whether you have received the required training
- ✓ Whether the equipment is suitable for the intended task
- ✓ Whether additional workplace controls apply

Key EU Product Directives

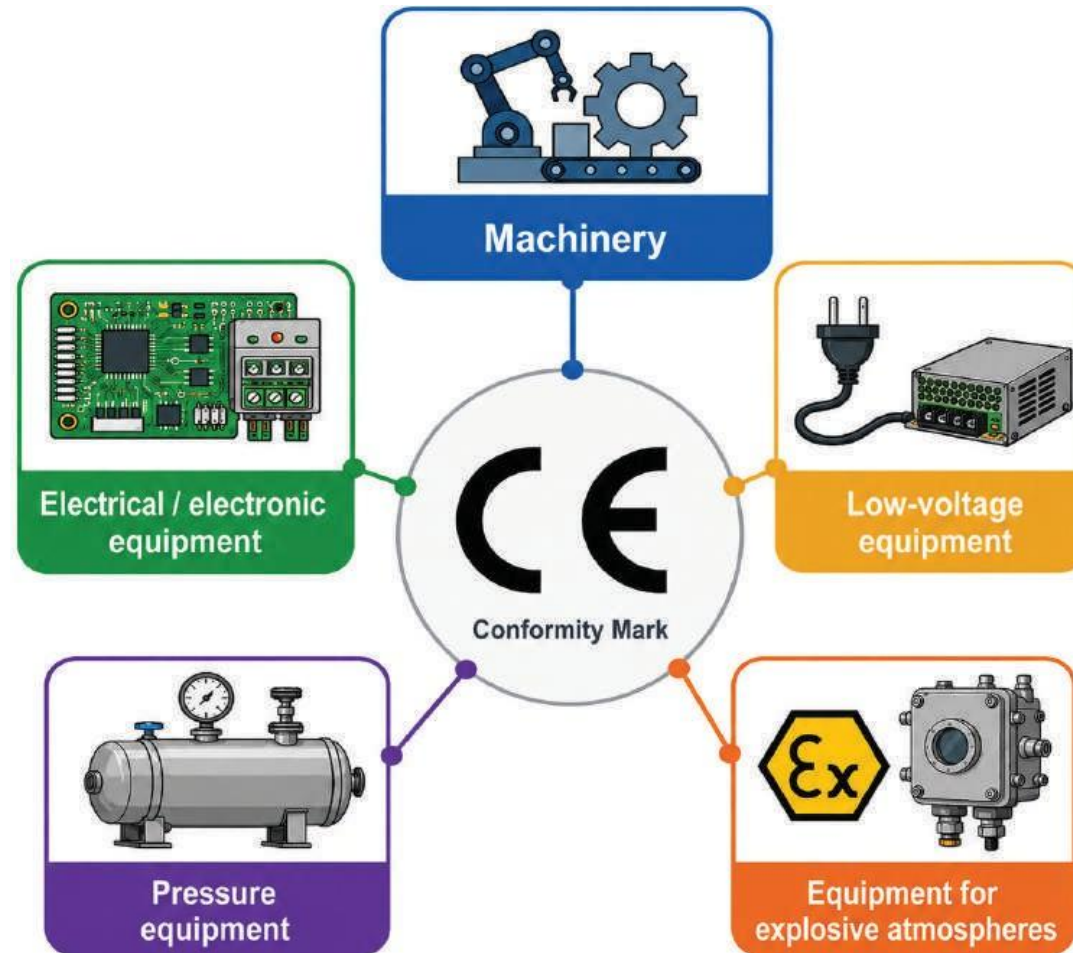
Before conformity can be declared, equipment is assessed against the EU directive that applies to its category — for example:

KEY PRODUCT-DIRECTIVE EXAMPLE

Directive	What it concerns	Example of relevance
Machinery Directive 2006/42/EC	Machinery containing connected components, with at least one moving part and a drive system other than direct human or animal effort	Production machinery, machine assemblies

Key EU Product Directives

Different equipment categories may be subject to different product directives



Different equipment categories may be subject to different product requirements

Partly Completed Machinery

Not every machine-related assembly is ready for independent operation.

The Machinery Directive recognizes partly completed machinery , a subassembly intended only to be incorporated into other machinery.

Partly completed machinery is accompanied by :-

- Assembly instructions.
- A Declaration of Incorporation — rather than the full Declaration of Conformity used for completed machinery
- This distinction matters because a component intended for incorporation into another machine must not automatically be treated as complete machinery ready for independent use.



(a)



(b)



(c)

Figure 3.3 — Examples of partly completed machinery: (a) servo valves; (b) rotary tables (fitted without safety devices); (c) hydraulic power units with solenoid valves.

The Ex Marking (ATEX Equipment)

Special requirements for equipment used in explosive atmospheres.



Some equipment is designed for use in areas where explosive atmospheres may be present .
The ATEX product requirements use a specific Ex marking in addition to the applicable CE marking.

An Ex marking **does NOT** mean "This equipment is explosion-proof in every situation".
It means special equipment requirements apply.

You must still follow :-

- ✓ The equipment information
- ✓ Manufacturer instructions
- ✓ Task-specific controls
- ✓ The authorised workplace classification
- ✓ Site procedures

Safety Critical: Your Role as a Technician



SAFETY CRITICAL: DO NOT MAKE YOUR OWN CONFORMITY DECISION

As a technician, do not:

- ✗ Certify equipment yourself
- ✗ alter an Ex marking
- ✗ add or reproduce a CE marking
- ✗ Declare an unmarked product compliant
- ✗ assume a similar-looking product has the same approval
- ✗ Release equipment for use because a marking appears to be present

Your role is to inspect the information provided, record what you observe, and report discrepancies through the authorized route.

Summary

Subtopic 3.3 in six points

1

Not a quality mark

CE marking is a conformity declaration, not a general quality mark.

2

Still check the basics

Condition, guards, instructions, training and task suitability all still matter.

3

Directives by category

Equipment falls under a specific EU directive, e.g. Machinery Directive 2006/42/EC.

4

Partly completed machinery

Needs a Declaration of Incorporation, not a full Declaration of Conformity.

5

Ex marking ≠ guaranteed safety

It means special equipment requirements apply — nothing more.

6

Never self-certify

Inspect the information provided, record what you observe, and report discrepancies.



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Unit 1 - Occupational Health, Safety and Environment

3.4 HOW TECHNICAL SAFETY STANDARDS SUPPORT INDUSTRIAL WORK

ENG – Nada Ahmed Ali Attia

Date/7/10/2026

Academic Year: 2026/2027
Semester : One

OBJECTIVES

BY THE END OF THIS LESSON, YOU WILL BE ABLE TO:

1

Define a Technical Standard

Explain what a technical standard is and distinguish it from a legally binding **Technical Regulation**.

3

Apply Standards to Technical Work

Understand how to recognize, evaluate, and apply standards referenced in workplace equipment documentation.

2

Classify Machinery Safety Standards (Types A, B1, B2, C)

Categorize standards based on their scope (from general design principles to specific machinery requirements).

4

Identify the Three Levels of Standardization

Distinguish between **International** (ISO, IEC), **European** (CEN, CENELEC, ETSI), and **National** standardization bodies.

Connecting Legal Principles to Engineering Practice



Definition - TECHNICAL STANDARD

A technical standard is a documented technical specification, guideline or assessment method developed through a recognized standardization process.



Purpose: Provides engineers with consistent technical methods for design, manufacturing, testing, inspection, and safety.



Three Levels of Standardization

The 3 Levels:

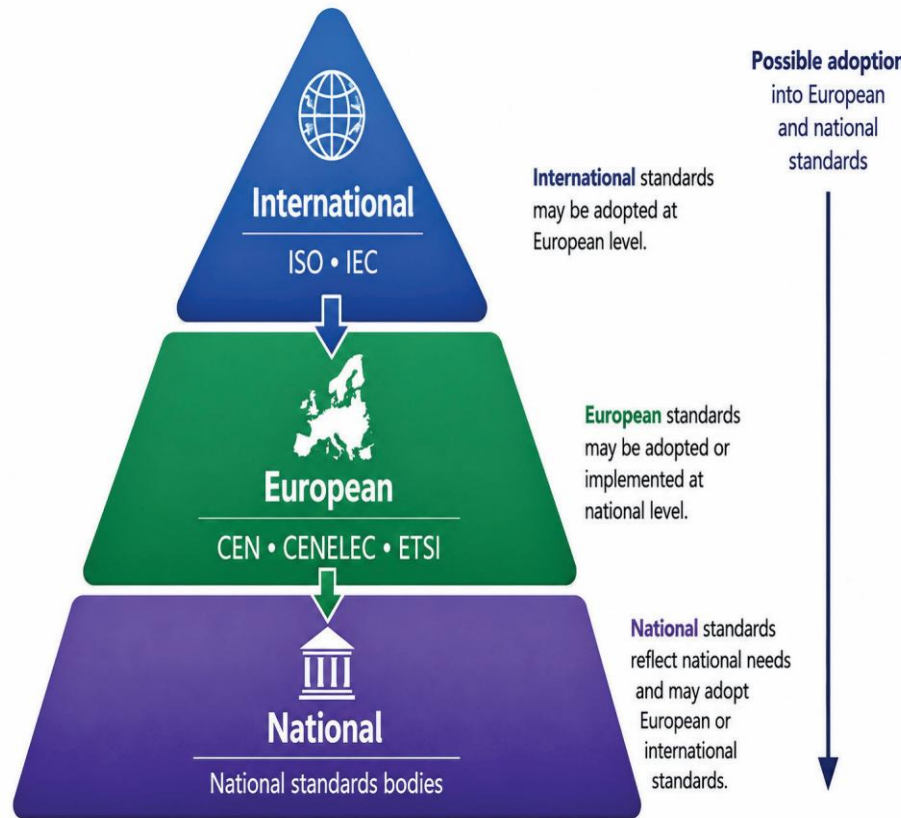
1.International Level: ISO (General) & **IEC** (Electrical/Electrotechnical).

2.European Level: CEN (General), **CENELEC** (Electrotechnical), and **ETSI** (Telecommunications).

3.National Level: Adoption of international and European standards within national legal/technical frameworks.

•**Takeaway:** The same standard often carries multiple identifiers showing adoption across all three levels (e.g., UNI EN ISO).

Three Levels of Standardization



Is a Technical Standard a Law?

Not automatically. A technical standard and a legal requirement are different

Technical Standard:

Voluntary by default; provides recognized technical methods.

Technical Regulation:

Legally binding requirement imposed by law.

When does a standard become mandatory?

- When referenced in legislation.
- When incorporated into technical regulations.
- When specified in binding organizational/contractual requirements.

Harmonised Machinery Safety Standards (Types A, B, C)

Classification Table:

Type	Scope	Description & Example
Type A	Basic Safety Principles	General design principles & risk assessment methodology (<i>EN ISO 12100</i>).
Type B1	Specific Safety Aspects	Covers safety aspects across various machines, e.g., control systems (<i>EN ISO 13849-1</i>).
Type B2	Specific Safety Devices	Covers protective devices, e.g., Emergency Stop (<i>EN ISO 13850</i>), safety distances (<i>EN ISO 13857</i>).
Type C	Specific Machine Category	Detailed requirements for a specific machine type, e.g., machine-tool presses (<i>EN ISO 16092-1</i>).

Priority Rule: *Type C standards take precedence over relevant Type A and B standards where requirements differ*

Homework & Class Exercises

Part A: Multiple Choice Questions

Which organization focuses specifically on electrical, electronic, and related technical standards at the international level?

- A) CEN
- B) IEC
- C) ETSI

What makes a Technical Regulation different from a Technical Standard?

- A) Technical standards are always mandatory laws.
- B) Technical regulations are legally binding, whereas standards are voluntary unless referenced by law.
- C) Technical regulations are published only by national companies.

Which type of machinery standard deals specifically with safety devices such as Emergency Stops (EN ISO 13850)?

- A) Type A
- B) Type B1
- C) Type B2
- D) Type C

If a Type C standard conflicts with a Type A standard for a specific machine category, which requirement takes precedence?

- A) The Type A standard
- B) The Type C standard
- C) The employer chooses whichever is easier



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Unit 1 - Occupational Health, Safety and Environment

3.5 READING MARKINGS, LABELS AND EQUIPMENT INFORMATION

ENG – Nada Ahmed Ali Attia

Date/7/10/2026

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Semester : One

EUROPEAN DIRECTIVES AND TECHNICAL SAFETY STANDARDS

READING MARKINGS, LABELS AND EQUIPMENT INFORMATION



EUROPEAN DIRECTIVES AND TECHNICAL SAFETY STANDARDS

OBJECTIVES

- Use safety **markings**, **labels** and **equipment documentation** to identify required precautions and user responsibilities.

EUROPEAN DIRECTIVES AND TECHNICAL SAFETY STANDARDS

READING MARKINGS, LABELS AND EQUIPMENT INFORMATION

PRODUCT IDENTIFICATION Check information such as

Manufacturer
name

model or type

serial or
identification



CHECK THE MARKINGS Look for markings that are relevant to the product.

CE marking

Ex marking where
applicable

warning symbols



EUROPEAN DIRECTIVES AND TECHNICAL SAFETY STANDARDS

READING MARKINGS, LABELS AND EQUIPMENT INFORMATION

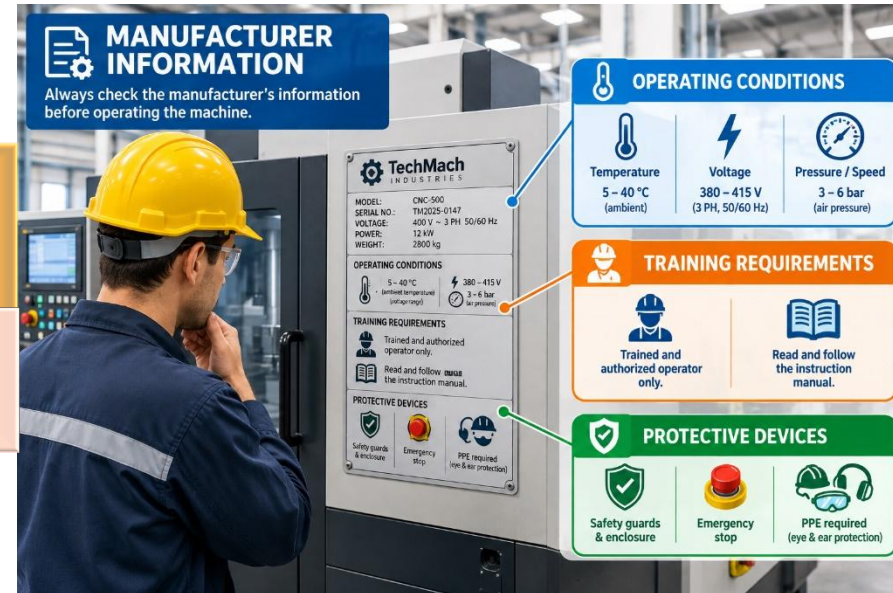
THE MANUFACTURER INFORMATION Manufacturer information may identify

- required operating conditions
- training requirements
- protective devices



DIFFERENT LABELS COMMUNICATE DIFFERENT INFORMATION

- pictograms
- H-statements
- P-statements



H-statement



H225

Highly flammable liquid and vapour.

Definition:

H-statements describe the nature and level of the hazard (what the chemical can do).

P-statement



P210

Keep away from heat, hot surfaces, sparks, open flames and other ignition sources. No smoking.

Definition:

P-statements tell you what precautions to take (how to prevent or reduce the risk).

EUROPEAN DIRECTIVES AND TECHNICAL SAFETY STANDARDS

READING MARKINGS, LABELS AND EQUIPMENT INFORMATION

INDUSTRIAL EXAMPLE

Situation: You are preparing to assist with an industrial machine. The equipment label shows:

Model: MX-200

Serial: 14582

CE marking: present

The operating
provided are labelled:

Model: MX-200A



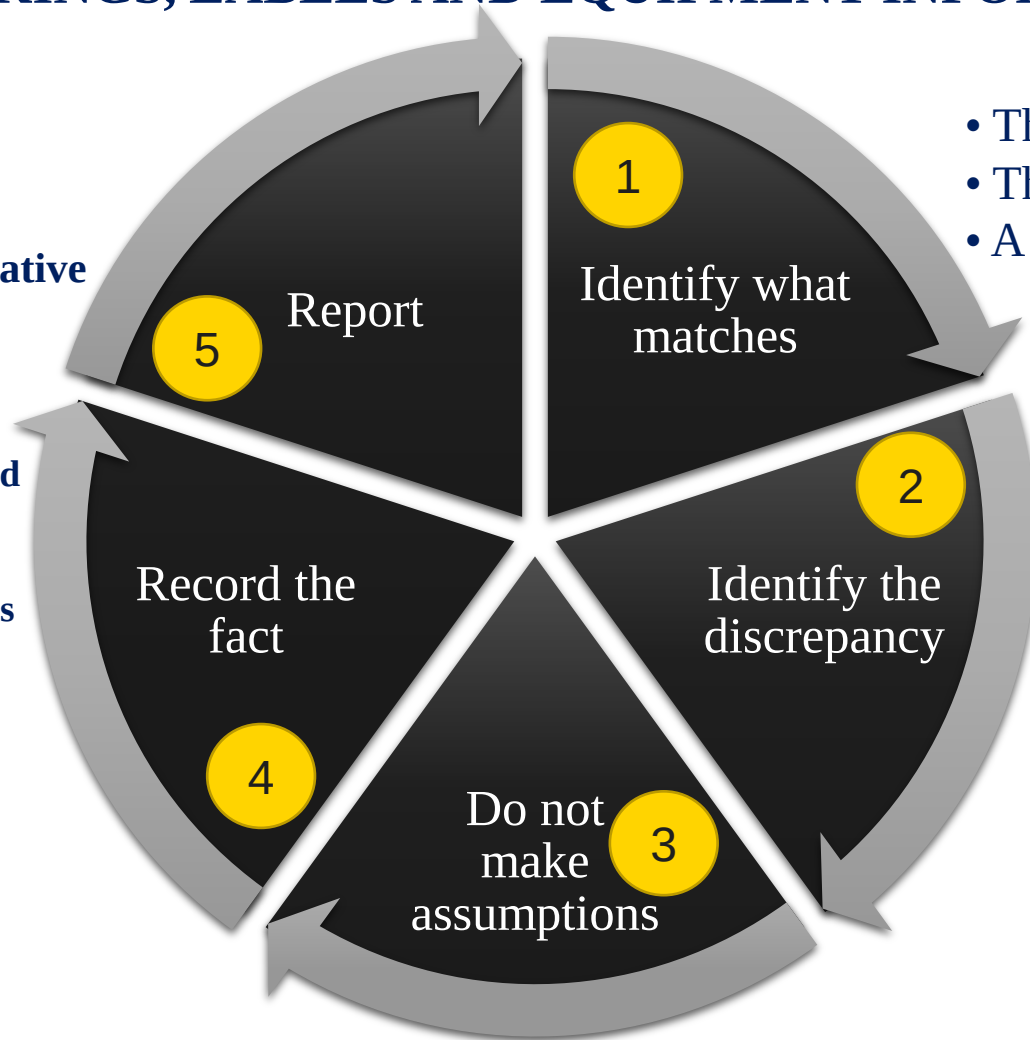
EUROPEAN DIRECTIVES AND TECHNICAL SAFETY STANDARDS

READING MARKINGS, LABELS AND EQUIPMENT INFORMATION

WHAT TO DO ?

- supervisor
- engineer
- safety representative

"The equipment is identified as Model MX-200. The supplied operating instructions are identified as Model MX-200A. Applicability of the instructions requires authorized confirmation before they are relied upon for the task."



- The model is **MX-200**.
- The serial number is **14582**.
- A CE marking is present

- Machine: **MX-200**
- Manual : **MX-200A**

- They look almost the same, so the manual must be correct, the machine is illegal

EUROPEAN DIRECTIVES AND TECHNICAL SAFETY STANDARDS

TEST YOUR UNDERSTANDING !

PART A — PUT THE PROCESS IN ORDER

Number these stages from 1 to 5.

Stage	Order
CE marking applied where required	5
Applicable EHSRs identified	2
Conformity assessment completed	3
Product directive identified	1
Declaration of Conformity prepared	4

EUROPEAN DIRECTIVES AND TECHNICAL SAFETY STANDARDS

TEST YOUR UNDERSTANDING !

PART B — CHECK YOUR KNOWLEDGE

1. Essential Health and Safety Requirements are:

- a. optional quality improvements
- b. essential health and safety requirements
- c. maintenance schedules
- d. worker training records

Answer:

2. Standards relating specifically to machinery safety devices are generally:

- a. Type A
- b. Type B1
- c. Type B2
- d. Type C

Answer:

3. The Electromagnetic Compatibility Directive 2014/30/EU is an EU product directive.

True ☐ False ☐



EUROPEAN DIRECTIVES AND TECHNICAL SAFETY STANDARDS

TEST YOUR UNDERSTANDING !

PART C — MATCH THE PRODUCT TO THE DIRECTIVE

Choose from:

Machinery • EMC • Low Voltage • Pressure Equipment • ATEX

Industrial item/situation	Directive
Pressure vessel operating above the specified pressure threshold	Pressure Equipment
Machine with powered moving parts	Machinery
Equipment intended for a potentially explosive atmosphere	ATEX
Electrical/electronic apparatus that must avoid unacceptable electromagnetic disturbance	EMC
Electrical equipment within the specified voltage range	Low Voltage

EUROPEAN DIRECTIVES AND TECHNICAL SAFETY STANDARDS

TEST YOUR UNDERSTANDING !

PART D — INTERPRET THE MARKING

1. What does CE marking communicate?

The manufacturer declares that the applicable EU requirements have been addressed and the required conformity-assessment procedures completed.

2. Does CE marking mean you can automatically use the equipment for any task?

☐ Yes

☐ No

Explain:

The manufacturer declares that the applicable EU requirements have been addressed and the required conformity-assessment procedures completed.

3. What does an Ex marking alert you to?

Special requirements apply to equipment intended for use in potentially explosive atmospheres.

EUROPEAN DIRECTIVES AND TECHNICAL SAFETY STANDARDS

TEST YOUR UNDERSTANDING !

4. A flame-over-circle pictogram appears on a chemical safety label.

What does it indicate?

- ☐ Flammable
- ☐ Oxidizing
- ☐ Pressurized gas
- ☐ Acute toxicity



EUROPEAN DIRECTIVES AND TECHNICAL SAFETY STANDARDS

TEST YOUR UNDERSTANDING !

You inspect the following:

Machine label:

Model: ST-450

Serial: 74218

Manufacturer instructions:

Model: ST-450

Serial reference: 74281

Answer:

1. What discrepancy do you identify?

The model matches, but the serial references do not.

2. Should you simply assume the document belongs to the machine? ☐ Yes ☐ No

3. Write a professional observation.

“The machine is identified as Model ST-450 Serial 74218. The supplied instructions refer to Model Applicability of the instructions requires authorized confirmation.”

4. What should happen next

Record and report the discrepancy for authorised document verification before relying on the instructions.

EUROPEAN DIRECTIVES AND TECHNICAL SAFETY STANDARDS

SUMMARY

- READING MARKINGS
- LABELS
- EQUIPMENT INFORMATION



EUROPEAN DIRECTIVES AND TECHNICAL SAFETY STANDARDS

HOMEWORK !

SECTION A: PAGE 175 SB

SAFETY FOUNDATIONS, RESPONSIBILITIES AND TECHNICAL REQUIREMENTS

QUESTION 4:

EQUIPMENT INFORMATION

A machine nameplate identifies the equipment as:

Model MX-200

The maintenance instructions supplied beside the machine are marked:

Model MX-200A

a. What discrepancy should you record?

.....

b. What should happen next?

.....